

Kevin McLeod

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EDUCATION

Harvard University

Graduation: May 2027

- **B.A. Double Major** in **Computer Science** and **Physics**, GPA: 3.65/4.0
- **Relevant Coursework:** Data Structures and Algorithms, Systems Programming & Machine Organization, Computing at Scale, Machine Learning, AI Planning & Learning Methods, Probability, Linear Algebra

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, C#, Java, SQL, HTML/CSS

Frameworks & Tools: React, Next.js, Node.js, FastAPI, Tailwind, Three.js / React Three Fiber, PyTorch, OpenCV, MediaPipe, Unity

Infrastructure & Tools: Postgres, Supabase, Celery, Git/GitHub, REST APIs, Linux

Systems & Concepts: Data Structures & Algorithms, Systems Programming, Operating Systems, Concurrency, Virtual Memory, Full-Stack Development, Databases, Data Pipelines, Machine Learning, Computer Vision, Multimodal Systems, GenAI Evaluation

WORK EXPERIENCE

Microsoft - Software Engineering Intern

May - August 2026

- **Shipped a new generative-AI image experience** on Bing end-to-end — design, implementation, offline evaluation, and went through the review process for shipping
- Prototyped intent-driven curated GenAI image experiences and a new routing design that curates for stacked user needs
- **Developed text-to-text quality metrics** to evaluate Bing image experiences and used them to improve multiple image experiences by **7%** and more

Stealth AI Startup - Co-founder & Engineering Lead (React, [Next.js](#), FastAPI, Tailwind, backend + AI models)

2025 - 2026

- **Ran discovery interviews and surveys with 100+ prospective users**, then narrowed scope based on findings
- Architected and built the full stack from an empty repo to a working prototype while iterating with user feedback

GP Enterprise Solutions - Full-Stack Software Engineering Intern (JavaScript, React, SQL)

June - August 2025

- **Optimized database response times by 5%** by restructuring filter logic and eliminating redundant queries, enabling faster access to high-volume financial data
- **Shipped client-requested features** such as bulk file export and dynamic form management, incorporating specific feedback from users through multiple interviews

Harvard AI Research Lab - Research Assistant with Professor Yilun Du (PyTorch)

May - August 2025

- Estimated relative 3D object positions from single RGB frames by pairing Mask R-CNN instance segmentation with MiDaS monocular depth estimation
- Built a PyTorch prototype composing per-object energy terms (ground contact, non-interpenetration, scene centering) into a differentiable scene-level objective minimized with Adam over 3D object positions

Harvard Datamatch - Web Developer (JavaScript, Next.js)

2024 - 2026

- **Rebuilt the Datamatch frontend** with a team of 5 people, serving 22,000+ students through a matching window

PROJECTS

ClipV.I.S. - TypeScript, Three.js, Node.js, MediaPipe

[Source Code](#)

- **Built real-time, webcam-based gesture controls** with MediaPipe and TypeScript, mapping pinch, point, and other gestures to 3D model selection, movement, rotation, and UI actions
- Integrated gesture recognition into a Three.js/React Three Fiber hologram experience that **won 3rd place** in a Microsoft Intern Hackathon category

Operating Systems Class Project - C++

[Source Code](#)

- **Developed a minimal OS from scratch** with a custom heap allocator, virtual memory, and syscall interface
- Built thread-safe I/O and a preemptive scheduler, simulating multitasking and safe concurrent file access
- **Reduced memory usage by 40%** via coalescing allocator with aligned block reuse and fragmentation mitigation

Calendar Syncing App - FastAPI, Postgres, Celery, Next.js/TypeScript

[Source Code](#)

- Built a **link-first group scheduling web app** where attendees respond to a shared link
- Integrated privacy-preserving Google Calendar sync with free/busy-only reads, ICS invites, and automated reminder jobs

Light in the Dark - C#, Unity

[Source Code](#)

- **Led a team of 10 people** on a 3D RPG game and directed cross-functional team coordination in game design, audio, and UI
- Built animation state machines, modular combat, dialogue, and cutscene systems using event-driven architecture

LEADERSHIP

Harvard Undergraduate Negotiation Club - Founding Board Member and Web Developer

January 2025 - Present

- **Designed and launched the club's [website](#)** using JavaScript and React; improved engagement for events and is now the primary channel for club member signups

FIRST Robotics Team 1493 - Head of Electrical and Programming, Pit Chief, and Outreach Lead

September 2019 - June 2023

- **Led electrical/programming** and outreach; Ranked 23rd at 2022 FIRST Robotics **World Championship** (Houston, TX)